

M.Feriansyah Raihan Taufiq *AI Engineer*



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📅 09/12/2000

🇮🇩 Indonesian

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🌐 Muhammad Feriansyah Raihan Taufiq

🔗 ML Engineer / Data Scientist Portfolio

🔗 AI Engineer Portfolio

👤 Profile

AI/ML Engineer (M.Sc. Computer Science, B.Sc. Physics — Universitas Indonesia) specializing in production LLM systems and computer vision. At PT. Indonesia Indicator I design and deploy multi-agent AI architectures, RAG pipelines, and MCP-based tooling across media intelligence, finance, security, and geopolitical research — covering everything from LLM evaluation (OpenAI, Claude, Gemini, self-hosted) to vector search, async orchestration, and deployment. My research roots are in deep learning and computer vision: both my undergraduate and graduate theses focused on generative models, image translation, and reconstruction. Core stack: Python, asyncio, PostgreSQL, Qdrant/vector DBs, Elasticsearch, Kafka, and Streamlit. I focus on building AI systems that are reliable in production, not just impressive in a demo.

🎓 Education

Master of Computer Science

Universitas Indonesia 📄

01/2023 – 01/2025

Depok, Indonesia

Graduate student majoring on Computer Science focusing on Artificial Intelligence with GPA 3,70/4,00.

Produce a thesis with the title "Improving Synthetic Nighttime Image Resolution in Day-to-Night Image Translation Based on Super-Resolution CycleGAN".

Bachelor of System & Instrumentation Physics

Universitas Indonesia 📄

09/2018 – 09/2022

Depok, Indonesia

Undergraduate student majoring Physics focusing on System & Instrumentation Physics with GPA 3,65/4,00 (Cumlaude).

Produce a thesis with the title "Hyperspectral Image Reconstruction from RGB Image Using Convolutional Neural Network with Dense Block Architecture: Case Study of Prediction System on (Diospyros Discolor Willd.) Leaf's Carotenoid Content".

Highschool Diploma in Natural Sciences

SMA Negeri 1 Pekanbaru 📄

2015 – 2018

Pekanbaru,
Indonesia

Professional Experience

Indonesia Indicator

Artificial Intelligence Researcher/Engineer

03/2025 – present
Tangerang Selatan,
Indonesia

Designing and deploying production-scale multi-agent AI systems across cross-functional teams, adapting architectures per client — media intelligence, security, finance, geopolitical research, and regional development.

- Built an OSINT social media profiling platform with 11+ agents for cross-platform account profiling — orchestrating social media API scraping, AI-powered web search, image analysis, and facial recognition for autonomous information gathering. Covers behavioral analysis, entity extraction, digital trust scoring, and credibility assessment.
- Developed a cognitive warfare detection system using causal inference to identify coordinated inauthentic behavior, proxy actor networks, and narrative manufacturing. Includes DFA manipulation detection and a three-tier campaign engine (Normal/Crisis/War) with velocity-based scoring.
- Built a Table Augmented Generation system (DocETL) — LLM pipelines that detect table structures, repair headers, resolve merged cells, and normalize Excel data. Exposed as MCP service.
- Built an equity analysis platform — DCF valuation, backtesting, and Streamlit screener dashboard.
- Designed 28 indicator agents for geoeconomic simulation with parallel country-watch analysis and MCP tool calling.
- Built the Stellar Deep Research Agent — a multi-phase research engine exposed as MCP service with web search and Wikipedia integration.
- Architected a Book Generator — auto-generates 80-100+ page feasibility studies with multi-model fallback and contextual RAG retrieval.
- Rapidly prototype functional PoC apps — AI command dashboards, fire intelligence platforms, and similar — to validate ideas and demo client-facing solutions quickly.
- Evaluating LLM models for local and cloud deployment across OpenAI, OpenRouter, and self-hosted setups.

Stack: Python, asyncio, Claude/GPT/Gemini, MCP, Qdrant, Vector DB, PostgreSQL, Elasticsearch, Kafka, Streamlit, S3

Fiverr Freelancer

Remote Data Scientist

11/2022 – 01/2025
Depok, Indonesia

A freelance tasked to work as a Data Scientist which build Machine/Deep Learning models.

- Consult with clients regarding the problems they want to overcome using a machine learning approach.
- Build Machine/Deep Learning models using python-based programming language.
- A final documentation on how to use the code will be reported to the client as well.

PPSDM Migas Cepu

06/2021 – 07/2021

Instrument Control Engineer Internship

Cepu, Indonesia

An internship to complete one of the physics subject focusing on System & Instrumentation Physics.

- Control the level parameter of Boiler Drum-Heat Exchanger BDT921 by choosing the right tuning PID methods.
- Obtain and analyze data by controlling level parameter of Boiler Drum-Heat Exchanger BDT921 with using a right tuning methods.
- Write a final report based on the collected data.

Electronics Laboratory Assistant

01/2021 – 12/2021

Head of Laboratory Assistant

Depok, Indonesia

A physics subject that aims to learn how to properly assemble analog or digital electronic circuits.

- Arrange the schedule of practicum about two semester at Department of Physics, University of Indonesia.
- Provide the latest idea about experiments on practicum with total 18 modules.
- Assist the electronic laboratory assistant if a student makes a mistake during the experiment.
- Giving final grades of electronics practicum subject to undergraduate physics students at the end of the semester.

**Volunteer Experience & Organisations****Himpunan Mahasiswa Departemen Fisika UI**

02/2019 – 09/2020

Staff of Human Resources Development & Entrepreneurship Department

Depok, Indonesia

Student organisation who works under Department of Physics, University of Indonesia which is tasked to facilitate and nurture physics students.

Human Resources Development:

- Monitoring internal activities of the organisation.
- Supervising organisation's event.

Entrepreneurship:

- Managing secondary income for the organisation.
- Selling organisation's merchandise

Pesta Rakyat Fisika

09/2018 – 11/2018

Staff of Public Relation, Public Lecture & Star Party

Depok, Indonesia

An annual event held by Himpunan Mahasiswa Departemen Fisika Universitas Indonesia which consist of science competition and participated by schools from all over Indonesia.

Public Relation:

- Informing schedule to high school student related to the competition.
- Become a liaison officer to maintain specific activities.

Public Lecture:

- Controlling the presentation.
- Monitoring the continuity of the event.

Star Party:

- Ensuring the equipment of the event.
- Monitoring weather forecast.

UI Goes to Riau 2019

Publications Editor

An annual event held by Ikatan Mahasiswa Riau Universitas Indonesia (IMR UI) which aims to introduce University of Indonesia to high school students from Riau.

- Gathering information from every faculty of University of Indonesia.
- Designing a brochure to promote University of Indonesia.

10/2018 – 02/2019

Pekanbaru,
Indonesia

Hard Skills

Programming Language

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Python, MATLAB, C

Google Cloud Platform

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Vertex AI, BigQuery ML, Machine Learning Operations (MLOps), Generative AI, LLM, Computer Vision, Model Deployment

AI Stack

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Context Engineering, Prompt Engineering, Retrieval Augmented Generation & Multi-Agent Systems

Big Data & Machine Learning

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Data Visualization, Exploratory Data Analysis, Data Engineering, Algorithm Modeling
(eg. scikit-learn for machine learning libraries & tensorflow 2.0 for deep learning libraries),
SQL (PostgreSQL)

Data Pipelines

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Apache Kafka, Airflow & Rees

Soft Skills

- Communication
- Teamwork
- Critical Thinking
- Time Management

Languages

Bahasa Indonesia

Native Speaker

English

Proficient

Certificates

- Python - Data Science (Sanbercode) 
- Advance Python - Data Science (Sanbercode) 
- Deep Learning Bootcamp (Data Science Center Universitas Indonesia) 
- The Complete SQL Bootcamp 2022: Go from Zero to Hero (Udemy) 
- Tensorflow 2.0: Deep Learning and Artificial Intelligence (Udemy) 